Electron-Muon Tree Level

X. Y

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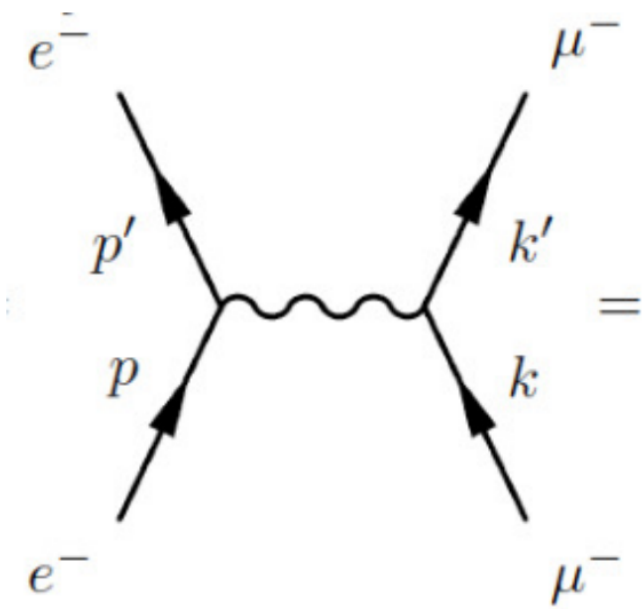
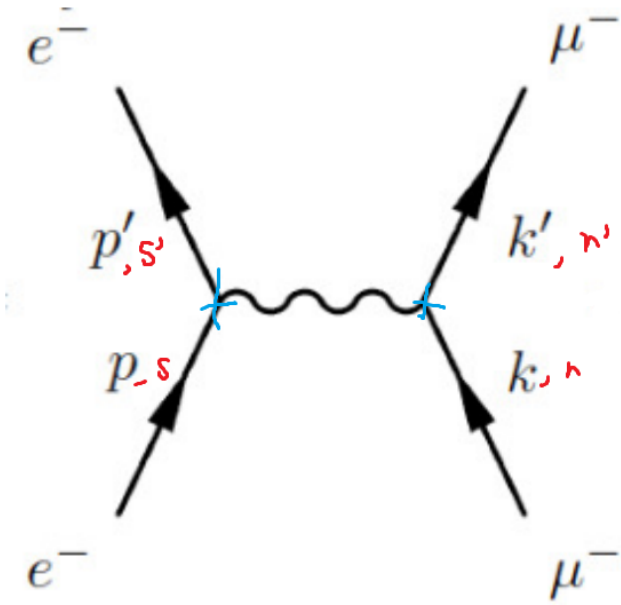
Correction to yanxi@tcd.ie or DM me via messenger.

(a) Calculate spin-averaged amplitude squared of the scattering of an electron with a muon ($\frac{1}{4}\sum|\mathcal{M}^2|$):



$$= i\mathcal{M}$$

Answer:
Let's first label the diagram



The diagram gives the following integral after applying Feynman rule:

$$
\begin{aligned}
i\mathcal{M}(e^-\mu^- \to e^-\mu^-) &= \bar{u}(p')(-ie\gamma^\mu)u(p)\frac{-i\eta^{\mu\nu}}{p^2_{\text{photon}}}\bar{u}(k')(-ie\gamma^\nu)u(k) \\
&= \frac{ie^2}{q^2}\bar{u}_{s'}(p')\gamma^\mu u_s(p)\eta^{\mu\nu}\bar{u}_{r'}(k')(\gamma^\nu)u_r(k)
\end{aligned}
$$

To evaluate

$$
\frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2
$$

We note that

$$
\begin{aligned}
\sum_{\text{spins}} u^s(p)\bar{u}^s(p) &\longrightarrow \not{p} + m \\
\sum_{\text{spins}} v^s(p')\bar{v}^s(p') &\longrightarrow \not{p}' - m
\end{aligned}
$$

For our case, we have 4

$$
\frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{1}{4}\frac{e^4}{p^4_{\text{photon}}}\text{tr}\Big[(\not{p}' + m_e)\gamma^\mu(\not{p} + m_e)\gamma^\nu\Big]\text{tr}\Big[(\not{k}' + m_\mu)\gamma_\mu(\not{k} + m_\mu)\gamma_\nu\Big]
$$

Using the property

$$
\begin{aligned}
\text{tr}[\gamma^\mu\gamma^\nu] &= 4\eta^{\mu\nu} \\
\text{tr}[\gamma^a\gamma^b\gamma^c\gamma^d] &= 4(\eta^{ab}\eta^{cd} - \eta^{ac}\eta^{bd} + \eta^{ad}\eta^{bc}) \\
\text{tr}[\#\text{odd } \gamma] &= 0
\end{aligned}
$$

Then

$$
\begin{aligned}
\text{tr}\Big[(\not{p}' + m_e)\gamma^\mu(\not{p} + m_e)\gamma^\nu\Big] &= \text{tr}[\gamma^a\gamma^\mu\gamma^b\gamma^\nu](p'_a p_b) + m_e^2\text{tr}[\gamma^\mu\gamma^\nu] \\
&= 4\Big(\eta^{a\mu}\eta^{b\nu} - \eta^{ab}\eta^{\mu\nu} + \eta^{a\nu}\eta^{\mu b}\Big)(p'_a p_b) + 4\eta^{\mu\nu}m_e^2 \\
&= 4\Big(p'^\mu p^\nu - p'_a p^a\eta^{\mu\nu} + p'^\nu p^\mu\Big) + 4\eta^{\mu\nu}m_e^2 \\
&= 4\Big(p'^\mu p^\nu + p'^\nu p^\mu - \eta^{\mu\nu}(p'\cdot p - m_e^2)\Big)
\end{aligned}
$$

and similarly

$$
\begin{aligned}
\mathrm{tr}\Big[(\not{k}' + m_\mu)\gamma_\mu(\not{k} + m_\mu)\gamma_\nu\Big] &= \mathrm{tr}\Big[\gamma_a\gamma_\mu\gamma_b\gamma_\nu k'^a k^b + m_\mu^2\gamma_\mu\gamma_\nu\Big] \\
&= \mathrm{tr}[\gamma_a\gamma_\mu\gamma_b\gamma_\nu](k'^a k^b) + \mathrm{tr}[\gamma_\mu\gamma_\nu](m_\mu)^2 \\
&= 4\Big(k'_\mu k_\nu - k'^a k_a \eta^{\mu\nu} + k'_\nu k_\mu\Big) + 4\eta^{\mu\nu}(m_\mu)^2 \\
&= 4\Big(k'_\mu k_\nu + k'_\nu k_\mu - \eta^{\mu\nu}(k'\cdot k - m_\mu^2)\Big)
\end{aligned}
$$

Thus

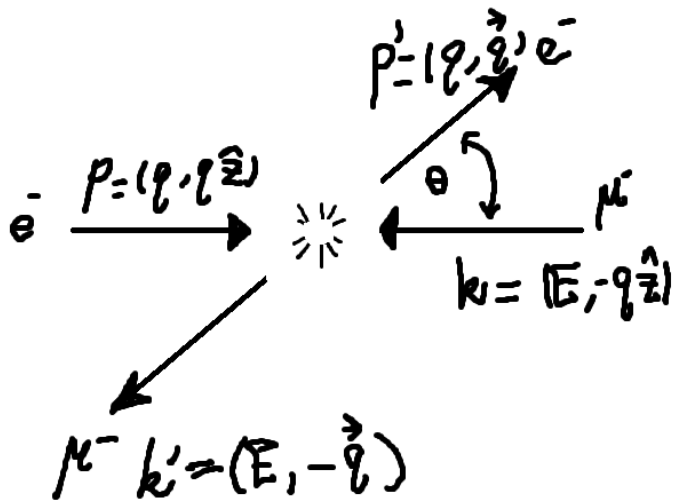
$$
\begin{aligned}
\frac{1}{4}\sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{1}{4}\frac{e^4}{p_{\text{photon}}^4}\mathrm{tr}\Big[(\not{p}' + m_e)\gamma^\mu(\not{p} + m_e)\gamma^\nu\Big]\mathrm{tr}\Big[(\not{k}' + m_\mu)\gamma_\mu(\not{k} + m_\mu)\gamma_\nu\Big] \\
&= \frac{1}{4}\frac{e^4}{p_{\text{photon}}^4}\Big\{4\Big(p'^\mu p^\nu + p'^\nu p^\mu - \eta^{\mu\nu}(p'\cdot p - m_e^2)\Big)4\Big(k'_\mu k_\nu + k'_\nu k_\mu - \eta^{\mu\nu}(k'\cdot k - m_\mu^2)\Big)\Big\} \\
&= 4\frac{e^4}{p_{\text{photon}}^4}\Big\{\Big(p'^\mu p^\nu + p'^\nu p^\mu - \eta^{\mu\nu}(p'\cdot p - m_e^2)\Big)\Big(k'_\mu k_\nu + k'_\nu k_\mu - \eta^{\mu\nu}(k'\cdot k - m_\mu^2)\Big)\Big\} \\
&= 4\frac{e^4}{p_{\text{photon}}^4}\Big(2(p'\cdot k')(p\cdot k) + 2(p\cdot k')(p'\cdot k) - 2(p'\cdot p)(k'\cdot k - m_\mu^2) - 2(k'\cdot k)(p'\cdot p - m_e^2) \\
&\qquad + (p'\cdot p - m_e^2)(k'\cdot k - m_\mu^2)\Big) \\
&= 4\frac{e^4}{p_{\text{photon}}^4}\Big(2(p'\cdot k')(p\cdot k) + 2(p\cdot k')(p'\cdot k) - 2(p'\cdot p)(k'\cdot k) + 2m_\mu^2(p'\cdot p) - 2(k'\cdot k)(p'\cdot p) \\
&\qquad + 2m_e^2(k'\cdot k) + 4(p'\cdot p)(k'\cdot k) - 4m_\mu^2(p'\cdot p) - 4m_e^2(k'\cdot k) + 4m_e^2 m_\mu^2\Big) \\
&= 4\frac{e^4}{p_{\text{photon}}^4}\Big(2(p'\cdot k')(p\cdot k) + 2(p\cdot k')(p'\cdot k) - 4(p'\cdot p)(k'\cdot k) + 2m_\mu^2(p'\cdot p) \\
&\qquad + 2m_e^2(k'\cdot k) + 4(p'\cdot p)(k'\cdot k) - 4m_\mu^2(p'\cdot p) - 4m_e^2(k'\cdot k) + 4m_e^2 m_\mu^2\Big) \\
&= 4\frac{e^4}{p_{\text{photon}}^4}\Big(2(p'\cdot k')(p\cdot k) + 2(p\cdot k')(p'\cdot k) - 2m_\mu^2(p'\cdot p) - 2m_e^2(k'\cdot k) + 4m_e^2 m_\mu^2\Big) \\
&= 8\frac{e^4}{p_{\text{photon}}^4}\Big((p'\cdot k')(p\cdot k) + (p\cdot k')(p'\cdot k) - m_\mu^2(p'\cdot p) - m_e^2(k'\cdot k) + 2m_e^2 m_\mu^2\Big)
\end{aligned}
$$

(b) Take the limit where muon mass m_μ is large and simplify the formula (limit when $\frac{m_e}{m_\mu} \to 0$)

Answer:
The limit is same as setting $m_e = 0$, which leads to

$$
\frac{1}{4}\sum_{\text{spins}} |\mathcal{M}|^2 = 8\frac{e^4}{q^4}\Big((p'\cdot k')(p\cdot k) + (p\cdot k')(p'\cdot k) - m_\mu^2(p'\cdot p)\Big)
$$

To evaluate the expression, we must work out the kinematics. Working in the center of mass frame. We make the following assignment



where

$$
\begin{aligned}
p &= (q, \vec{q}\hat{z}) \\
p' &= (q, \vec{q}) \\
k &= (E, -\vec{q}\hat{z}) \\
k' &= (E, -\vec{q})
\end{aligned}
$$

$$
\begin{aligned}
E^2 &= q^2 + m_\mu^2 \\
q \cdot \hat{z} &= q\cos\theta \\
E_{\text{cm}} &= E + q \\
p_{\text{photon}}^2 &= (p - p')^2 = -2p \cdot p' \\
&= -2((q)(q) - qq\cos\theta) = -2q^2(1 - \cos\theta)
\end{aligned}
$$

Thus

$$
\begin{aligned}
p \cdot k = p' \cdot k' &= q(E + q) \\
p \cdot k' = p' \cdot k &= q(E + q\cos\theta)
\end{aligned}
$$

Thus our expression now becomes

$$
\begin{aligned}
\frac{1}{4}\sum_{\text{spins}} |\mathcal{M}|^2 &= 8\frac{e^4}{p_{\text{photon}}^4}\Big((p' \cdot k')(p \cdot k) + (p \cdot k')(p' \cdot k) - m_\mu^2(p' \cdot p)\Big) \\
&= 8\frac{e^4}{(-2q^2(1 - \cos\theta))^2}\Big(q^2(E + q)^2 + q^2(E + q\cos\theta)^2 - m_\mu^2(q)q(1 - \cos\theta)\Big) \\
&= \frac{2e^4}{q^2(1 - \cos\theta)^2}\Big((E + q)^2 + (E + q\cos\theta)^2 - m_\mu^2(1 - \cos\theta)\Big)
\end{aligned}
$$